Rice - Neck blast

RAG record ADV-008 | canonical key: rice_neck_blast

Record ID	ADV-008
Crop / crop key	Rice / rice
Condition / disease key	Neck blast / neck_blast
Main pathogen	Magnaporthe oryzae (syn. Pyricularia oryzae)
RAG aliases	rice neck blast, panicle blast, rice blast, Magnaporthe oryzae, neck rot, white panicle

Farmer quick guide: treatments protect healthy plant tissue and slow disease spread. Dead or badly damaged leaf tissue will not become green again.

How to recognize it

- The panicle neck becomes grey-brown or dark and may be girdled.
- If infection happens early, grain may not form; later infection can cause poorly filled or low-quality grain.
- The panicle may droop or appear white, but white panicles can also be caused by stem borer or water stress.

Weather or field conditions that favour it

High relative humidity and wet weather around heading favour neck blast.

What to do now

- Check the panicle neck closely before assuming every white panicle is blast.
- Avoid excessive nitrogen and keep scouting from booting through heading.
- When neck blast risk is high or first symptoms are seen, use a currently registered option promptly.

Chemical treatment - only after the disease is confirmed

Chemical / formulation	Rate from cited source	How to use this information	Source
Tricyclazole	0.6 g/L water	ICAR advisory option for neck blast.	[S1]

Chemical / formulation	Rate from cited source	How to use this information	Source
Isoprothiolane	1.5 ml/L water	ICAR advisory alternative for neck blast.	[S1]
Kasugamycin	2.5 ml/L water	ICAR advisory alternative for neck blast.	[S1]

Safety and compliance: Use only a product whose current Indian label allows the crop and target disease. Follow the label rate, water volume, re-entry interval and pre-harvest interval. Wear label-required PPE. Do not mix products unless the label or a qualified extension adviser permits it.

Prevention and integrated management

- Use blast-resistant varieties where available.
- Avoid excessive nitrogen.
- Reduce carry-over infection with clean seed and crop sanitation.
- Time protection around booting/heading according to local blast advisories.

RAG guardrail

The AI should treat the chemical table as source-grounded examples, not permission to invent a different pesticide or dose. If the retrieved source does not match the crop, disease and formulation, return non-chemical advice and request local verification.

Sources used for this record

[S1] Indian Council of Agricultural Research (ICAR). ICAR crop advisory - neck blast management.
https://icar.gov.in/sites/default/files/2022-06/eBook-COVID-19-ICAR-Initiatives.pdf